

Foundation Mathematics 1017SCG
Week 7 Workshop Questions

1. Sketch each of the following equations. Your graph should clearly show the x -intercept and y -intercept. Remember to label your axes.

- (a) $y = x - 2$
- (b) $y = -x + 3$
- (c) $y = 2x - 6$
- (d) $y = -2x + 4$
- (e) $y = 3x + 12$
- (f) $y = -3x + 3$
- (g) $y = 4x - 8$
- (h) $y = -4x$

2. Find the equation of the line with the following properties. Write your equation in the form $y = mx + c$ where m and c are constants.

- (a) Passes through $(-1, 4)$ and has a gradient of 3.
- (b) Passes through $(2, -3)$ and has a gradient of -1 .
- (c) Passes through $(-5, 2)$ and has a gradient of 4.
- (d) Passes through $(-3, 1)$ and $(4, 15)$.
- (e) Passes through $(-4, 0)$ and $(0, 3)$.
- (f) Passes through $(1, 1)$ and $(5, 17)$
- (g) Has an x -intercept at $x = 2$ and a y -intercept at $y = 6$.

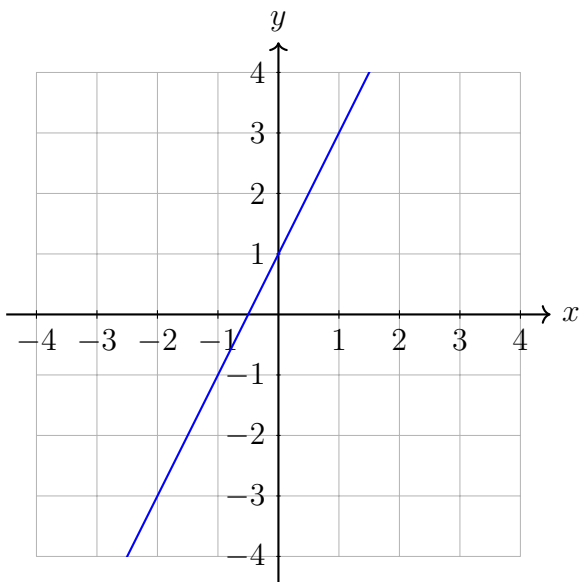
Hint: If the line has an x -intercept at $x = 2$, what point does the line pass through?

- (h) Has an x -intercept at $x = -5$ and a y -intercept at $y = -5$.

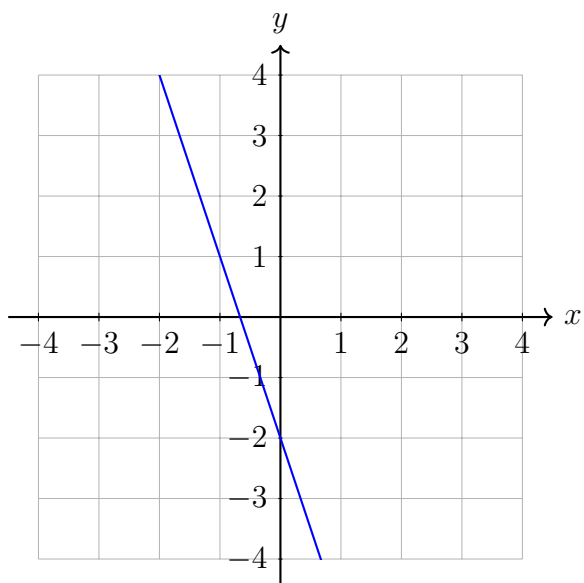
3. Find an equation for each of the following graphs in the form $y = mx + c$ where m and c are constants.

Hint: If you can determine two points that the graph passes through, you can complete this question in a similar way to Question 2.

- (a)



(b)



4. Sketch each of the following functions. Your graph should clearly show the turning point, y -intercept and any x -intercepts. Remember to label your axes.

(a) $y = x^2 - 2x - 3$

(b) $y = -x^2 + 2x + 8$

(c) $y = x^2 - 9$

(d) $y = -2x^2 - 12x - 10$

(e) $y = x^2 - 8x + 16$

(f) $y = x^2 - 4x + 7$

5. Solve the following simultaneous equations.

Hint: The simultaneous equations in this question have integer solutions.

(a)

$$3x + 2y = 7$$

$$5x - y = 3$$

(b)

$$3x + 5y = -5$$

$$4x - y = 24$$

(c)

$$4x + 3y = -9$$

$$3x + 2y = -7$$

(d)

$$5x - 4y = 10$$

$$4x - 3y = 7$$

(e)

$$5x + 2y = -4$$

$$7x + 3y = -6$$

(f)

$$4x - 3y = -19$$

$$10x - 2y = -42$$